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Paleomagnetic study of Upper Jurassic rocks from the Sichuan basin: tectonic aspects for the collision between the Yangtze Block and the North China Block

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Abstract

A paleomagnetic study has been carried out on Late Jurassic redbeds in the Sichuan basin, at the northern part of the Yangtze Block. Upper Jurassic brownish-red sandstones and brownish-red siltstones were collected in the Penglaizhen Formation at 26 sampling sites around Jiangyang city (30.4°N, 104.5°E). Thermal demagnetization isolated a high temperature magnetization component with a maximum unblocking temperature of about 690°C. The primary nature of magnetization acquisition is ascertained through a 95% confidence level positive fold test, as well as a positive reversal test. The tilt-corrected overall mean direction of 24 sites is $D = 25.9°$, $I = 26.7°$ ($\alpha_{95} = 4.7°$), leading to a paleomagnetic pole situated at 61.3°N/222.7°E ($A_{95} = 4.2°$). This pole position agrees with two other Late Jurassic poles from the Sichuan basin, indicating that this basin has behaved as a rigid block since Late Jurassic. These three poles from the Sichuan basin therefore provide a characteristic Late Jurassic pole for the stable part of the Yangtze Block (64.3°N, 231.2°E, $A_{95} = 6.7°$). Comparison of the Late Jurassic pole of the Yangtze Block with the contemporaneous pole for the North China Block (NCB) identifies a significant difference, implying that the relative movement between the Yangtze Block and the NCB continued during, and probably after, the Late Jurassic. Paleomagnetic and geological evidence reveals that the Yangtze Block collided first with the NCB during the Early Permian, and continued to penetrate into it while undergoing a clockwise rotation until the Early Cretaceous. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; tectonics; Upper Jurassic; Yangtze Platform; plate collision

1. Introduction

The Asian continent is a mosaic of tectonic blocks (Fig. 1a). One of the most prominent tectonic events that affected these blocks during the Mesozoic is the collision between the Yangtze Block and the NCB. This collision occurred along the east-trending Qinling–Dabieshan suture zone. According to paleomagnetic studies [1,2], the two blocks first collided during the Permian near their eastern parts, and then progressively scissored together as the Yangtze Block rotated clockwise with respect to the NCB. This scissor-like colli-

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sion model is supported by the progressive westward transition from marine to continental sedimentation along the Qinling–Dabieshan suture zone [3].

The age assignment for the cessation of collision is controversial, although a scissor-like motion of the collision is confirmed by both geological and paleomagnetic evidence. The geological estimate leads to an age older than that estimated paleomagnetically. Stratigraphic correlations and sedimentary features within the Qinling–Dabieshan suture zone suggest a Late Triassic cessation of the collision [4,5], whereas preliminary paleomagnetic studies claim that the collision continued until the Middle Jurassic [6] or even the Middle Jurassic–Late Jurassic boundary [7–9]. This discrepancy on the age assignment is due to different definitions for the cessation of the collision. Indeed, the geological evidence reveals the timing of closure of a proto-ocean between the Yangtze Block and the NCB, whereas the paleomagnetic data provide an age for the cessation of the relative motion some time after the ocean that existed between them had vanished. In order to decipher the relative motion between these two blocks at the latest stages of collision after the closure of the Qinling–Dabieshan suture zone, detailed post-Late Triassic paleomagnetic studies are required for the Yangtze Block as well as for the NCB.

Post-Late Triassic paleomagnetic poles have been computed for areas spread over the Yangtze Block [2,7–9]. The northwestern and southwestern parts of this block were subjected to Cenozoic intra-cratonic deformation [10–12], and paleomagnetic poles from these areas can not be regarded as representative of the stable block. It is therefore necessary to establish a characteristic paleomagnetic pole from a tectonically stable part of the Yangtze Block, in order to understand its relationship to the NCB.

We have focused in this study on Late Jurassic formations from the Sichuan basin that is considered to be a stable part of the Yangtze Block. Indeed, this basin is widely covered by Jurassic and Cretaceous strata in which no faults are observed. On the basis of Late Jurassic paleomagnetic data from this area, a characteristic Late Jurassic pole for the Sichuan basin is established, and compared to a pole for the NCB in order to examine relative motions between the two blocks.

2. Geological setting and sampling

The Sichuan basin extends for 1000 km E–W and 500 km N–S, and is mainly filled with Mesozoic continental redbeds (Fig. 1b). According to the Bureau of Geology and Mineral Resources of Sichuan Province [13], Upper Jurassic strata in the Sichuan basin are subdivided into the Suining and the Penglaizhen Formations, ascendingly. The Penglaizhen Formation consists of sandstones and mudstones. It underlies the Lower Cretaceous Chengqiannyan Group disconformably. The Penglaizhen Formation is assigned to the Late Jurassic on the basis of abundant ostracods such as *Darwinula oblonga* and *Darwinula leguminella* [14].

Paleomagnetic sampling was carried out around Jiangyang city in the northwestern part of the Sichuan basin, and focused on the Upper Jurassic Penglaizhen Formation. In the northwestern part of this basin, Upper Jurassic and Lower Cretaceous redbeds are gently folded along northeast-trending axis. As early Neogene Formations unconformably overly Jurassic to Paleogene strata in the northwestern margin of the basin [15], the folding is considered as having occurred at the early Neogene.

The Penglaizhen Formation widely crops out as a monoclinal sequence with dip shallower than 10° south of Jiangyang city, whereas northwest of the city this formation is folded into a northeast-trending anticline (Fig. 2). Twenty-six sites were sampled, eight of which were collected from both limbs of the anticline (three sites from the northern limb and five sites from the southern limb). The remaining 18 sites were collected from the monoclinal sequence. Only one sedimentary layer was sampled at each site. These sedimentary layers consist of brownish-red-colored rocks (siltstones, silt-sandstones and fine-grained sandstones), except for three red silt-sandstone layers (JY16, JY17 and JY18). Five to ten hand samples were collected at each site. The present declination value at the sampling site is

estimated from the 1995 International Geomagnetic Reference Field [16].

3. Paleomagnetic results

Samples were cut into 23-mm long and 25-mm diameter specimens. Remanent magnetization was measured using a 2G cryogenic magnetometer. Stepwise thermal demagnetization up to 690°C was conducted with a Natsuhara TDS-1 thermal demagnetizer. Acquisition of isothermal remanent magnetization (IRM) and back-field demagnetization were performed on a 2G pulse magnetizer up to maximum fields of 2.7 T. Magnetic components were resolved using principal component analysis [17].

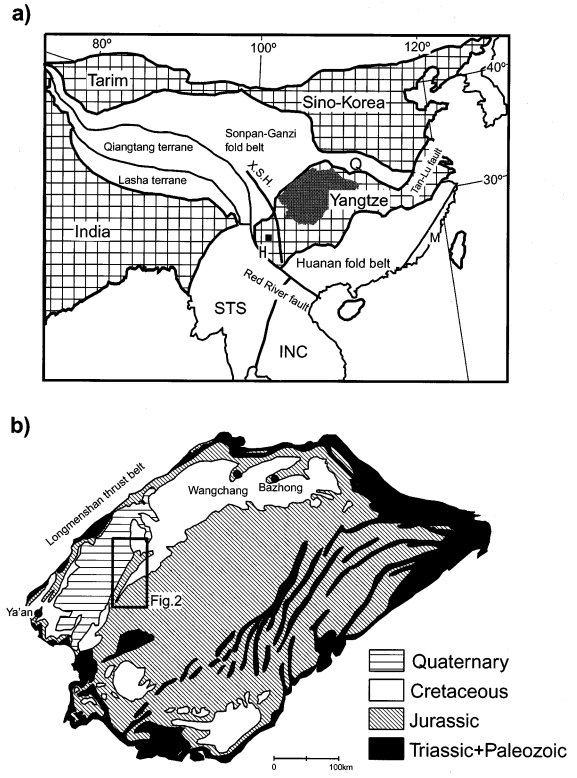


Fig. 1. (a) Simplified tectonic map of East Aisa (based on Zhang et al. [28]). Shaded area shows the Sichuan basin. H = location of Huili; STS = Shan-Thai and Simao Terrane; INC = Indochina Terrane; Q = Qinling accretionary fold belt; M = Maritime fold belt; X.S.H. = Xian-shui-he Fault. (b) Geological map of the Sichuan Basin. The box encloses the sampling area. Based on the geological map of Sichuan, scale 1:1 000 000 [13].

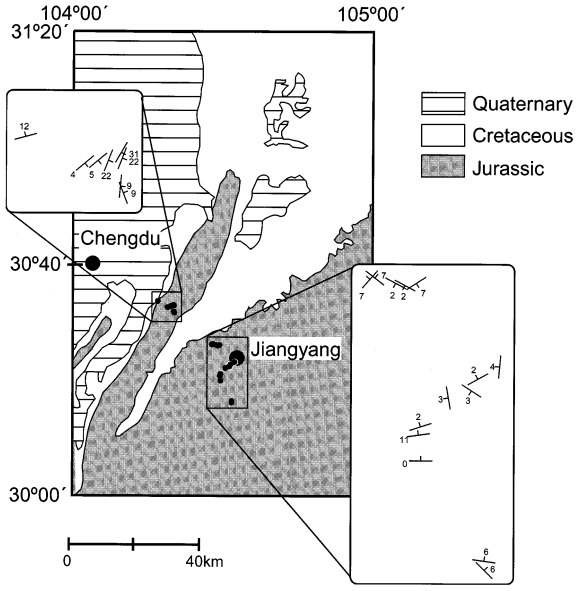


Fig. 2. Geological sketch map of the sampling area at Jiangyang. Solid circles indicate sampling sites. Bedding strike and dip at each sampling site are shown in the insets.

For the brownish-red-colored specimens, the larger grain sizes possess the weaker natural remanent magnetization (NRM) intensities [$3.3 \pm 2.6(10^{-3}$ A/m) for siltstones, $2.2 \pm 1.5(10^{-3}$ A/m) for silt-sandstones, and $1.6 \pm 0.8(10^{-3}$ A/m) for fine-grained sandstones]. Thermal demagnetization revealed two magnetic components. The low temperature component (LTC) was erased by 150°C. The high temperature component (HTC), with an unblocking temperature of 690°C, appeared above 450°C for the siltstones (Fig. 3a,b), whereas the silt-sandstones and fine-grained sandstones revealed distinct HTC in the temperature range between 650 and 690°C (Fig. 3c,d). No HTC was isolated from the fine-grained sandstones in one site (JY06).

Specimens from the red silt-sandstones have a stronger magnetization than those from the brownish-red silt-sandstones [the mean NRM intensity is $6.5 \pm 5.0(10^{-3}$ A/m)]. These specimens yielded two magnetic components during thermal demagnetization (Fig. 3e). After cleaning the LTC in the first stages of demagnetization (by 300°C),

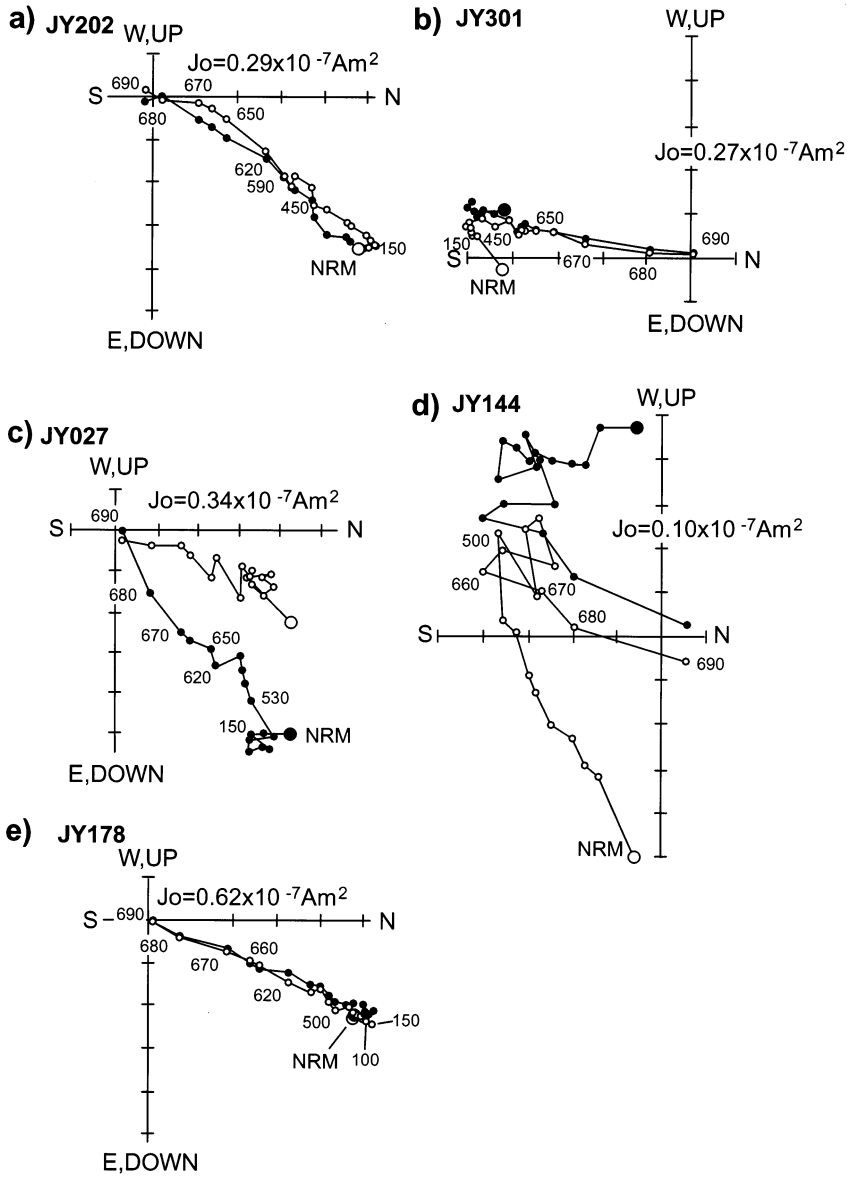


Fig. 3. Thermal demagnetization diagrams (in situ coordinate). Temperature steps are in °C. Orthogonal projection of demagnetization for (a, b) brownish-red siltstones (site JY20 and JY30), (c) brownish-red silt-sandstones (JY02), (d) brownish-red fine-grained sandstones (JY14) and (e) red silt-sandstones (JY17).

the remanent magnetization decayed linearly to the origin; this is reached at 690°C, suggesting that magnetization is carried by hematite.

The presence of hematite is confirmed by the IRM acquisition and back-field demagnetization experiments (Fig. 4). Brownish-red-colored and red silt-sandstones specimens show that no saturation is achieved after applying a field of 2.7 T and that the coercivity of remanence is about 0.4 T.

The LTC in situ directions are dominated by a component in the direction of the present Earth's field. Their mean direction at the sample level being $D = 354.4°$, $I = 50.7°$ ($\alpha_{95} = 7.5°$), is parallel

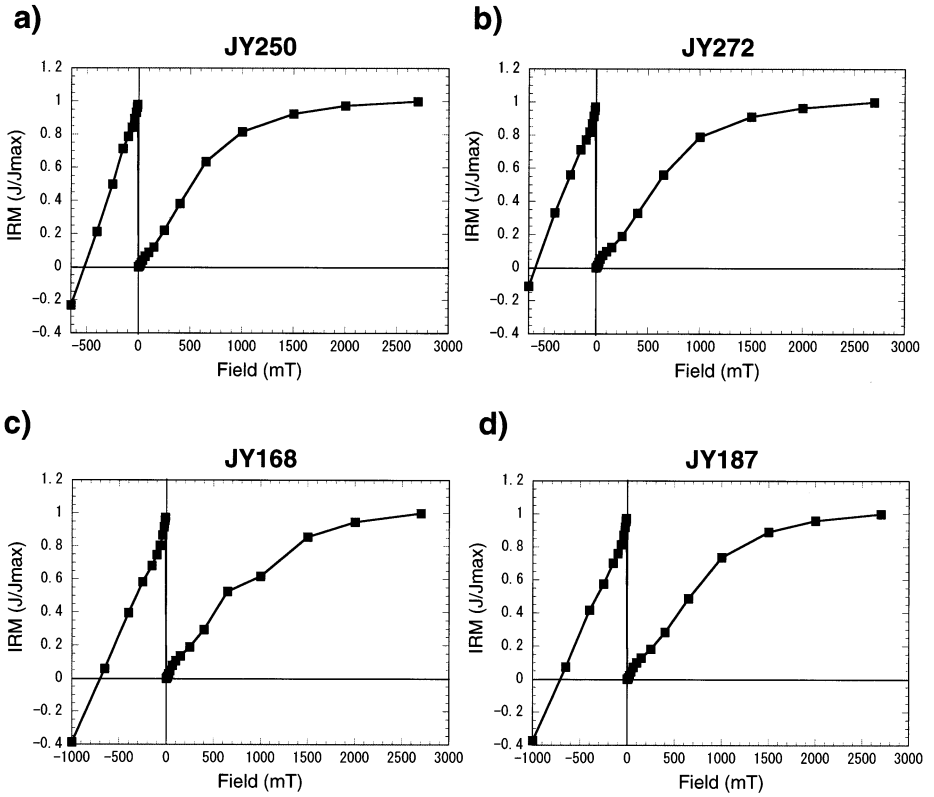


Fig. 4. Normalized IRM acquisition and back-field demagnetization curves. (a) Brownish-red siltstones (JY24). (b) Brownish-red silt-sandstone (JY27). (c, d) Red silt-sandstone (JY16 and JY18).

to the present Earth's field direction at the Jiangyang area ($D = 358.5°$, $I = 45.8°$).

If we focus on the HTC, each site contains specimens with only one polarity. Paleomagnetic directions, calculated from the directions of the HTC's, are obtained from 24 out of 26 sites (Table 1 and Fig. 5). Normal polarities are encountered at 13 sites and reversed ones at 11 sites. No paleomagnetic direction could be calculated for site JY07 because of the large scatter.

A formation mean direction is calculated from the 24 sites that display the HTC and is $D = 24.9°$, $I = 28.8°$ ($k = 35.8$, $\alpha_{95} = 5.0°$) in in situ coordinates, and $D = 25.9°$, $I = 26.7°$ ($k = 40.1$, $\alpha_{95} = 4.7°$) in stratigraphic coordinates.

4. Reliability of paleomagnetic directions

Because the paleomagnetic directions for eight sites come from both limbs of an anticline, fold tests can be applied. The mean direction from the eight sites is $D = 20.1°$, $I = 28.8°$ ($k = 36.4$, $\alpha_{95} = 9.3°$) in in situ coordinates, and $D = 23.1°$, $I = 26.8°$ ($k = 67.0$ $\alpha_{95} = 6.8°$) in stratigraphic coordinates. The precision parameter (k) increases from 36.4 to 67.0 after tilt correction, although the fold test of McElhinny [18] is inconclusive at the 95% confidence level. A rigorous fold test proposed by McFadden and Jones [19] is positive at the 95% confidence level; the statistic P is 0.677 in in situ and 0.086 in stratigraphic coordinates, while the critical value is 0.648 at the 95% confidence level. The tilt-corrected mean direction is thus considered to be of pre-folding origin.

A possible pre-folding origin for the paleomagnetic directions from the remaining 16 sites sampled in the monoclinal sequence is examined by comparing the tilt-corrected mean directions

Table 1
Paleomagnetic results for the Upper Jurassic redbeds at Jianyang

Site	Latitude	Longitude	*n*/*N*	In situ		Tilt-corrected		*k*	α_{95}	Polarity	Lithology
				Dec.	Inc.	Dec.	Inc.				
	(°N)	(°E)		(°)	(°)	(°)	(°)		(°)		
Layers on the anticline											
Northern limb											
JY21	30.55	104.28	8/9	220.0	−32.9	214.5	−25.8	13.7	15.5	R	fine-grained sandstone
JY22	30.55	104.28	7/8	207.0	−39.5	201.4	−30.3	10.4	19.7	R	fine-grained sandstone
JY23	30.55	104.28	6/9	202.8	−27.8	199.7	−18.3	22.3	14.5	R	fine-grained sandstone
Southern limb											
JY27	30.54	104.34	4/7	16.8	33.1	19.1	35.2	23.6	19.3	N	siltstone
JY28	30.54	104.34	6/8	12.8	40.0	17.0	40.3	19.1	15.7	N	silt-sandstone
JY29	30.52	104.34	8/9	194.2	−21.3	203.0	−21.6	20.1	12.6	R	fine-grained sandstone
JY30	30.52	104.34	6/8	198.9	−16.1	205.1	−15.2	24.4	13.8	R	siltstone
JY31	30.52	104.34	4/8	182.4	−16.3	194.3	−27.2	130.4	8.1	R	siltstone
mean	30.50	104.30	8/8	20.1	28.8			36.4	9.3		
						23.1	26.8	67.0	6.8		
Layers on monoclinal sequence											
JY01	30.27	104.53	3/7	218.0	−28.5	218.2	−20.6	22.4	26.7	R	fine-grained sandstone
JY02	30.27	104.53	6/8	42.6	26.0	39.1	29.4	19.4	15.6	N	silt-sandstone
JY03	30.26	104.53	8/9	54.9	23.1	53.2	19.0	24.4	11.4	N	silt-sandstone
JY06	30.35	104.49	0/6	–	–	–	–	–	–	–	fine-grained sandstone
JY07*	30.35	104.49	3/7	214.5	−18.7	213.8	−17.2	7.1	50.1	–	fine-grained sandstone
JY08	30.36	104.51	4/4	26.4	30.8	25.0	32.6	17.3	22.8	N	siltstone
JY11	30.37	104.53	4/5	23.2	41.0	21.8	39.7	21.9	20.1	N	siltstone
JY12	30.38	104.54	5/7	23.7	34.7	21.0	36.1	15.4	20.1	N	siltstone
JY14	30.43	104.47	3/8	217.5	−20.4	214.9	−19.9	70.9	14.8	R	fine-grained sandstone
JY15	30.43	104.47	4/8	213.7	−22.7	213.1	−19.7	11.9	27.8	R	fine-grained sandstone
JY16	30.43	104.48	8/8	15.1	28.6	16.4	21.5	57.8	7.3	N	silt-sandstone
JY17	30.43	104.48	7/8	20.7	24.7	21.5	17.5	79.9	6.8	N	silt-sandstone
JY18	30.43	104.49	8/9	14.0	22.8	13.7	25.3	45.4	8.3	N	silt-sandstone
JY19	30.43	104.49	8/8	14.7	22.1	14.5	24.6	27.5	10.7	N	siltstone
JY20	30.43	104.49	8/8	24.0	39.5	29.3	42.9	61.7	7.1	N	siltstone
JY24	30.54	104.32	4/10	201.7	−12.6	201.6	−6.5	25.5	18.6	R	silt-sandstone
JY25	30.54	104.33	9/9	186.8	−36.2	191.7	−31.7	13.4	14.6	R	siltstone
JY26	30.54	104.33	6/8	37.3	36.5	41.9	31.5	47.9	9.8	N	silt-sandstone
Mean	30.40	104.50	16/18	27.3	28.7			35.8	6.3		
						27.4	26.6	32.8	6.5		
Mean											
All	30.40	104.5	24/26	24.9	28.8			35.8	5.0		
						25.9	26.7	40.1	4.7		
Normal			13/26	25.3	31.6			39.4	6.7		
						25.8	31.0	36.6	6.9		
Reserved			11/26	204.5	−25.4			32.9	8.1		
						206.0	−21.7	59.1	6.0		

The magnetization directions of HTC are listed. *N* and *n* are numbers of specimens in each sites and specimens used for statistics, respectively. Dec. = declination; Inc. = inclination; *k* = precision parameter; α_{95} = radius of the 95% confidence circle. One site (denoted by *) is discarded because the circle of 95% confidence is large ($\alpha_{95} = 50.1°$); the other sites have the circle of 95% confidence less than 27.8°.

from the monoclinal sequence and from the anticline. The former is $D = 27.4°$, $I = 26.6°$ ($\alpha_{95} = 6.5°$), whereas the latter is $D = 23.1°$, $I = 26.8°$ ($\alpha_{95} = 6.8°$). These two directions are statistically indistinguishable at the 95% confidence level (the angular distance is $3.8° \pm 9.4°$). We thus conclude that the tilt-corrected mean direction from the monoclinal sequence was acquired

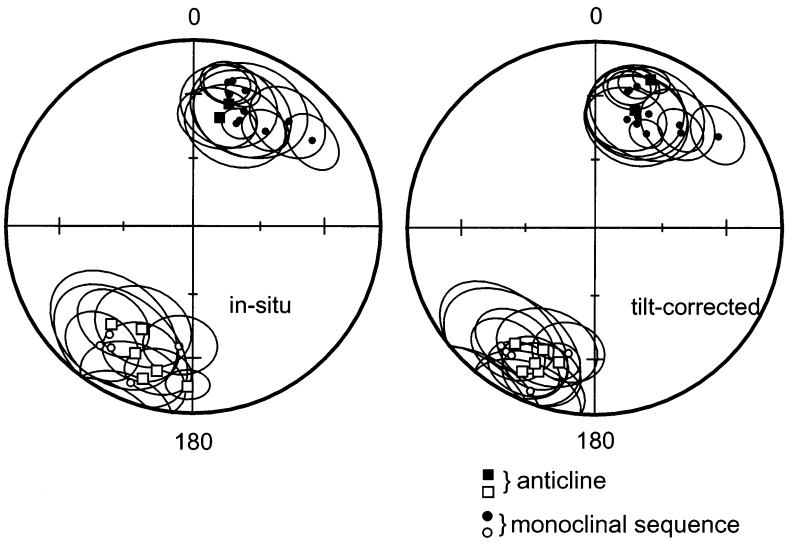


Fig. 5. Equal-area projection of the paleomagnetic directions of 24 sites before and after tilt correction. The mean direction is $D = 24.9°$, $I = 28.8°$ ($k = 35.8$, $\alpha_{95} = 5.0°$) in in situ coordinates, and $D = 25.9°$, $I = 26.7°$ ($k = 40.1$, $\alpha_{95} = 4.7°$) in stratigraphic coordinates. Paleomagnetic direction for each site is drawn with its circle of confidence at 95% level; solid (open) symbols on the lower (upper) hemisphere.

when the pre-folding magnetization was acquired in the anticline.

The tilt-corrected normal and reversed polarity mean directions deviate slightly from antipodality. The angular distance between them is of 9.3°, whilst the critical angle is of 9.0° [20]. This means that the tilt-corrected mean directions of normal and reversed polarities pass the reversal test as 'B' classification [20]. The overall tilt-corrected mean direction of the 24 sites therefore represents the paleomagnetic direction for a period of time in the Late Jurassic, embracing at least two field polarities.

5. The characteristic Late Jurassic pole for the stable part of the Yangtze Block

The Late Jurassic paleomagnetic pole for the Jiangyang area lies at 61.3°N/222.7°E ($A_{95} = 4.2°$) (Table 2). This pole is compared to other coeval poles for the Sichuan basin in Table 2 and Fig. 6a. Late Jurassic poles based on reliable paleomagnetic data sets with a positive fold test are reported from Bazhong [12] and Wangchang [21], and lie respectively at 64.7°N, 236.0°E ($A_{95} = 7.0°$) and 66.6°N, 236.4°E ($A_{95} = 7.6°$). While the Jiangyang pole is computed for the southern area of the northwestern part of the Sichuan basin, those of Bazhong and Wangchang counties are for its northern area. However, the pole for Jiangyang is indistinguishable from the poles of Bazhong and Wangchang at the 95% confidence level ($6.9° \pm 8.2°$ and $8.0° \pm 8.7°$, respectively). This indicates that no significant relative motion has occurred between the northern and southern areas of the northwestern part of the Sichuan basin since the Late Jurassic.

The pole for Jiangyang is compared with the coeval pole for Ya'an (29.9°N, 102.9°E) in the northwestern peripheral part of the Sichuan basin [10]. The pole for Jiangyang is significantly different from the pole for Ya'an at the 95% confidence level ($19.5° \pm 14.2°$). This difference can be described as a significant counter-clockwise rotation ($19.7° \pm 15.1°$) and a non-significant southward displacement ($5.3° \pm 11.4°$) of the Ya'an area. This indicates that Ya'an area suffered relative rotation with respect to Jiangyang area, and to the other area of the Sichuan basin in general.

Table 2
Mesozoic paleomagnetic poles from the North China Block and the Yangtze Block

Age	Locality	Latitude	Longitude	Pole position			N	Reference
		(°N)	(°E)	Latitude (°N)	Longitude (°E)	A_{95}, dp/dm (°)		
The North China Block								
K1	Ordos	35.0	108.0	75.8	208.7	7.5	10	[29]
J3	Anhui	31.6	116.0	74.4	222.8	5.9	10	[9]
J3	Shandong(1)	35.7	117.9	77.8	235.9	6.9	5	[24]
J3	Shandong(2)	36.6	117.9	73.5	207.8	5.7	6	[24]
J3	Mean			75.5	220.8	6.4	3	
J2	Anhui	31.6	116.0	72.9	254.7	6.4	17	[9]
J2	Yan'an	36.7	109.2	74.3	232.8	4.5	6	[6]
J1	Fuxian	36.2	109.3	82.4	286.0	7.0	10	[6]
T3	Tongchuan	35.2	109.1	62.3	7.7	3.8	11	[30]
T2	Tongchuan	35.2	109.1	59.8	359.9	4.0	17	[30]
T1	Hancheng	35.5	110.2	61.9	348.3	16.0	16	[30]
T1	Taiyuan	37.8	112.3	50.0	351.0	6.0	25#	[23]
P2	Taiyuan(1)	37.8	112.3	44.3	357.8	6.9	12	[31]
P2	Taiyuan(2)	37.8	112.3	47.8	357.4	4.0	89#	[23]
The stable part of the Yangtze craton								
K1	Guanyin	29.1	104.6	77.6	218.8	8.5	10	[10]
J3	Jiangyang	30.4	104.5	61.3	222.7	4.2	24	this study
J3	Bazhong	31.8	106.7	64.7	236.0	7.0	11	[12]
J3	Wangchang	32.1	106.2	66.6	236.4	7.6	6	[21]
J3	Mean			64.3	231.2	6.7	3	
T1	Tongzi	28.6	106.9	46.3	219.1	10.9	9	[32]
T1	Lianghekou	29.6	103.4	56.2	227.6	1.7/3.4	101#	[33]
T1	Chongqing	29.2	107.4	44.3	217.4	3.5	88#	[34]
T1	Hechuan	30.0	106.3	38.5	209.8	1.7/3.3	62#	[35]
T1	Hechuan	30.0	106.3	38.7	211.8	3.2/6.3	25#	[35]
T1	Mean			44.9	216.4	8.2	5	
P2	Emeishan	29.6	103.4	53.7	287.9	3.1/6.2	22	[31]
P2	Emaishan	29.6	103.4	54.5	254.3	1.8/3.5	24	[33]
P2	Emaishan	29.6	103.4	57.0	260.9	4.3/8.5	12#	[33]
P2	Mean			56.0	267.9	15.9	3	

P2 = Late Permian; T1 = Early Triassic; T2 = Middle Triassic; T3 = Late Triassic; J1 = Early Jurassic; J2 = Middle Jurassic; J3 = Late Jurassic; K2 = Early Cretaceous; A_{95} = radius of the 95% confidence circle; dp (dm) = small (great) axis of the ellipse of confidence of the paleopole; N = number of sites; # = number of samples.

For this reason, we decide not to include the Ya'an pole in the computation of the Late Jurassic characteristic pole for the Sichuan basin.

A mean Late Jurassic pole for the Sichuan basin has thus been calculated using poles from the Bazhong, Jiangyang and Wangchang localities that cover a large area of the northwestern part of the basin; this pole lies at 64.3°N/231.2°E (A_{95} = 6.7°). In the central part of this basin, Upper Jurassic strata widely crop out as flat-lying sequences, indicating that the central part of the basin has experienced little tectonic deformation since the Late Jurassic. The Bureau of Geology and Mineral Resources of Sichuan Province [13] report that no large faults affect the Upper Jurassic strata between the northwestern and central parts of the basin. Based on this stable aspect, pointed out by geological evidence as well as by similar pole positions from the three areas, the Sichuan basin is assumed in the following as a tectonically stable area. Therefore, we consider that the pole we have computed represents a characteristic Late Jurassic pole for the tectonically stable part of the Yangtze Block.

The mean pole computed for the rigid Yangtze Block is significantly different from the Ya'an

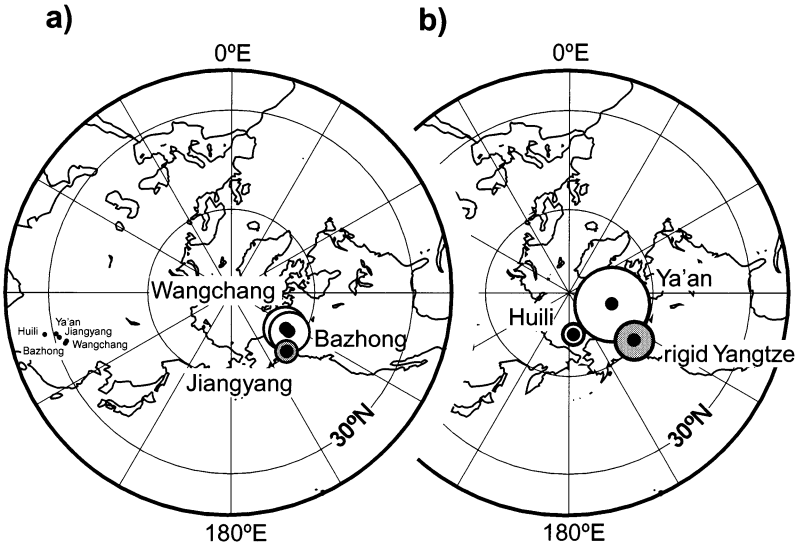


Fig. 6. Late Jurassic paleomagnetic poles from the Yangtze Block. (a) Paleomagnetic poles from the stable part of the block. (b) Mean paleomagnetic pole for the stable part of the block as well as poles from Ya'an and Huili. The pole for Ya'an lies at 77.9°N/255.4°E (A_{95} = 13.6°), while the pole for Huili at 74.4°N/185.1°E (A_{95} = 3.9°). Radius of the 95% confidence circle of Ya'an pole has been recalculated.

pole at the 95% confidence level (15.4° ± 15.2°) (Fig. 6b), confirming that a significant relative motion has occurred between these two areas. As the Ya'an area is situated within the Longmenshan thrust belt that separates the Sichuan basin from the Songpan–Ganzi fold belt, we suggest that this relative motion is due to local scale deformation following thrusting along the Longmenshan thrust belt.

The mean pole we have computed is significantly different from the pole of Huili [7,8] derived from the southwestern part of the Yangtze Block (Fig. 6b). The discrepancy between these pole positions suggests a southward movement of the Huili area with respect to the rigid Yangtze Block. Yokoyama et al. [12] attribute this movement to a large southward displacement of the southwestern part of the Yangtze Block as an independent tectonic block, referred to as the 'Kungdian High' [22]. The amount of southward displacement evaluated from the present study is 16.0° ± 8.2°, which corresponds to 1800 ± 900 km. The pole from the southwestern part of the Yangtze Block has therefore been an excluded from calculation of the characteristic pole for the stable part of this block.

6. Tectonic aspects of the collision between the Yangtze Block and the NCB

In order to discuss tectonic aspects of the collision between the Yangtze Block and the NCB, the paleomagnetic poles of these two blocks are compared during the interval between the Late Permian and Early Cretaceous (Table 2, Fig. 7).

Since the Sichuan basin is now identified as a tectonically stable part of the Yangtze Block, new characteristic Late Permian and Early Triassic poles for the Yangtze Block are recalculated from paleomagnetic poles reported from the Sichuan basin. The Early Cretaceous pole is recalculated from paleomagnetic directions of four sites at Guanyin (29.1°N, 104.6°E) and four sites at Xinjin (30.4°N, 103.8°E) [10]. No Middle Triassic to Middle Jurassic poles are computed because of the lack of paleomagnetic data from the Sichuan basin for that time window.

Late Permian to Early Cretaceous poles for the NCB are based on reviews by Enkin et al. [2] and Gilder and Courtillot [9]. Permo–Triassic magnetostratigraphic data recently reported [23] are accepted as reliable, and listed in Table 2. The characteristic Late Jurassic pole is computed using

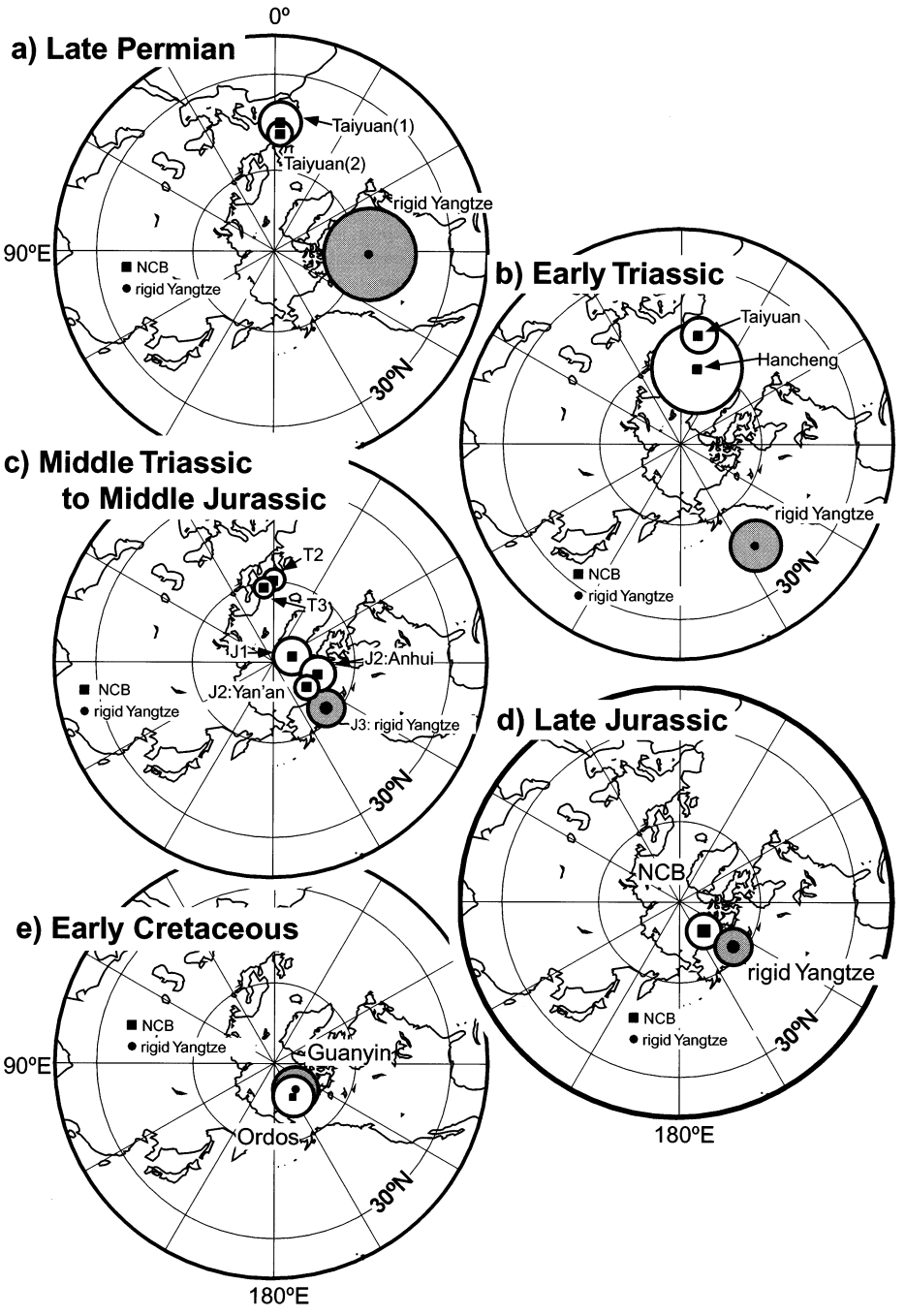


Fig. 7. Mesozoic paleomagnetic poles for the Yangtze Block and the NCB.

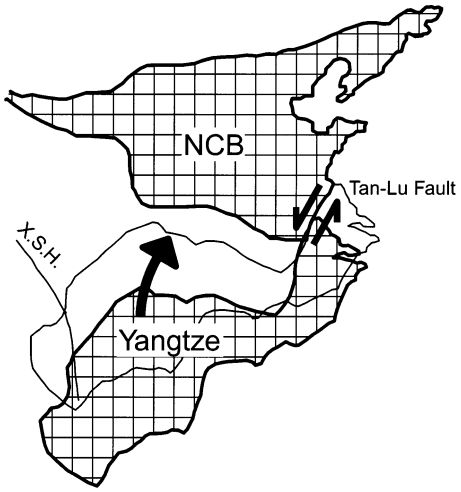


Fig. 8. A synthetic model of the collision between the Yangtze Block and the NCB from the Late Jurassic to the present. The present position of the Yangtze Block is shown in thin lines.

three poles: two poles from Shandong and one pole from Anhui [9,24].

The poles of the Yangtze Block are located apart from those of the NCB at the Late Permian and the Early Triassic (Fig. 7a,b). The angular distances are larger than 55°. The pole positions for the NCB from the Middle Triassic to the Middle Jurassic successively approach the Late Jurassic pole position of the Yangtze Block (Fig. 7c). By the Late Jurassic, the pole position of the Yangtze Block is still significantly different from that of the NCB at the 95% confidence level ($11.7° \pm 9.2°$) (Fig. 7d). The paleomagnetic poles from the two blocks, however, occupy the same position by the Early Cretaceous (Fig. 7e). We suggest on the basis of these results that the relative motion between the Yangtze Block and the NCB was still taking place during the Late Jurassic. The scissor-like collision phenomena which occurred later than the Early Permian [1] has most likely been taking place until the Early Cretaceous.

The relative motion of the Yangtze Block with respect to the NCB, between the Late Jurassic and the Early Cretaceous, is described in terms of clockwise rotation ($6.8° \pm 9.8°$) and northward displacement ($9.8° \pm 7.4°$) of the Yangtze Block (Fig. 8). The Euler pole, which superimposes the poles of the Yangtze Block and NCB, lies at approximately 20.0°N/150.0°E. As the solution for a Euler pole is not unique, we have chosen the one situated east of the Yangtze Block to enable both the clockwise rotation and northward displacement of the Yangtze Block to take place. The displacement about this Euler pole predicts a left-lateral motion along the Tan–Lu Fault system that represents the eastern boundary between the Yangtze Block and the NCB. According to Xu et al. [25], a left-lateral displacement of 1000–1500 km took place along this fault system during the Mesozoic. The highest peak of this left-lateral displacement occurred in the last part of the Late Jurassic or the earlier part of the Early Cretaceous between 140 Ma and 130 Ma (the average rate of this left-lateral displacement is up to 5–10 cm/yr). The sense and timing of the movements on the Tan–Lu Fault system, determined by field observation, is consistent with those expected from paleomagnetism. The clockwise rotation and northward displacement of the Yangtze Block is in our opinion a realistic Late Jurassic tectonic scenario for the collision.

Since geological investigation of the Qinling–Dabieshan suture zone indicates that ocean closure was complete by the Late Triassic [3–4], the Late Jurassic relative motion between these blocks is attributable to penetration of the Yangtze Block into the NCB. Okay and Sengor [26] proposed that exhumation of the ultra-high-pressure rocks within the Qinling–Dabieshan suture zone is mainly caused by intracontinental thrusting and concomitant with erosion and back thrusting of this suture zone. They proposed that this phenomenon is similar to that in the Himalayas that resulted from the indentation of India. The intracontinental thrusting would have given rise to a Late Jurassic to Early Cretaceous heat source, which is suggested by U–Pb ages (132 ± 10 Ma and 152 ± 13 Ma) on granites in the Qinling–Dabieshan suture zone and in the Shandong area [27]. We conclude that penetration of the Yangtze Block is a Late Jurassic tectonic aspect of the collision between the Yangtze Block and the NCB.

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